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# UNIT 7 DETERMINANTS

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## 7.0 INTRODUCTION

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As you know linear algebra, a fundamental field of mathematics that serves as the bedrock for modern data science, engineering, physics, and computer graphics. At the core of this discipline one very important concept is the **determinant** which has deep structural and computational significance. In this unit you will learn the understanding, calculating, and applying determinants, moving systematically from their fundamental definition to one of their most advanced applications in solving simultaneous equations and analysing matrix structure. A determinant, typically denoted as  $\det(A)$  or  $|A|$ , is a unique scalar value associated only with a square matrix (a matrix with the same number of rows and columns). Historically, the concept arose in the 17th century from the problem of solving systems of linear equations. Mathematically, the determinant reveals whether a matrix is invertible (non-singular) or not. To efficiently compute the determinant of matrices larger than  $3 \times 3$ , we must employ a structured, recursive method. This is where the concepts of Minors and Cofactors become indispensable. This unit will provide a brief of the cofactor expansion method, demonstrating how it breaks down an  $n \times n$  determinant calculation into a sum of determinants of  $(n-1) \times (n-1)$  matrices, forming the primary computational tool for the entire field.

Calculating large determinants directly using the definition can be computationally intensive and highly susceptible to error. Therefore, a core section of this chapter is dedicated to the Properties of Determinants. These properties are not just theoretical curiosities; they are efficiency engines that allow for substantial simplification. This unit will demonstrate how operations such as row or column interchange, multiplying a row by a scalar, and adding a multiple of one row to another (elementary row operations) affect the determinant's value in a predictable way. Also, in this unit you will learn how to calculate the Product of Two Determinants, and find the Adjoint of a matrix, which is closely linked to the matrix inverse.

Matrices often possess specific structural symmetries that simplify their analysis. In this unit we will explore Symmetric Determinants and Skew-symmetric Determinants

Finally, towards end of the unit, we will focus on the most direct use of the determinant: solving systems of linear equations. **Cramer's Rule** provides a formula for Determinant-based method for finding the unique solution to a system of  $n$  equations with  $n$  variables, provided the coefficient matrix is non-singular.

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## 7.1 OBJECTIVES

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Determinants were first used in solving systems of simultaneous equations in around 1683 almost simultaneously by Japanese mathematician Seki Kowa and German mathematician Leibnitz. The approach adopted by them, for introducing the concept of determinant, is almost the same as is explained in the very beginning of next section. According to renowned mathematician L. Debnath, historically, determinants were discovered over two centuries before the discovery of matrices.

### Learning Objectives

After going through this unit, you should be able to:

- explain the concept of determinant and various mathematical operations on determinants;
- use the properties of determinants;
- find product of two determinants; and
- solve a system of simultaneous linear equations by Cramer's rule.

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## 7.2 DETERMINANTS

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Let us start our discussion with definition and concepts of determinants.

### 7.2.1 Definition and Concept

If two linear equations in one variable viz.  $a_1x + b_1 = 0$  and  $a_2x + b_2 = 0$ , with  $a_1$ ,  $a_2$ ,  $b_1$ , and  $b_2$ , here being assumed as non-zeroes, have the same/common solution,

then  $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ , i. e.,  $\frac{a_1}{b_1} = \frac{a_2}{b_2}$ , i. e.,  $a_1b_2 - a_2b_1 = 0$ . The expression  $(a_1b_2 - a_2b_1)$  is called a **determinant** and is denoted by the symbol  $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ .

Note that,  $a_1$ ,  $a_2$ ,  $b_1$ ,  $b_2$  are called the '**elements of the determinant**'. The elements may be zeroes also. The elements in the horizontal direction form '**rows**' and those in the vertical direction form '**columns**'. The determinant  $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$  has two rows and two columns. So, it is called a '**determinant of the second order**' and it has  $2! = 2$  terms in its expansion, Again, if the three linear equations in two variables, viz.

$$\begin{aligned} a_1x + b_1y + c_1 &= 0, \\ a_2x + b_2y + c_2 &= 0 \text{ and} \end{aligned}$$

$$a_3 x + b_3 y + c_3 = 0$$

have common solution, then, by cross multiplication, from the last two equations., we get

$$\frac{x}{b_2 c_3 - b_3 c_2} = \frac{y}{c_2 a_3 - c_3 a_2} = \frac{1}{a_2 b_3 - a_3 b_2}$$

$$\text{or, } x = \frac{b_2 c_3 - b_3 c_2}{a_2 b_3 - a_3 b_2}, y = \frac{c_2 a_3 - c_3 a_2}{a_2 b_3 - a_3 b_2}$$

where  $(a_2 b_3 - a_3 b_2) \neq 0$

For the solution to be common, these values of x and y must satisfy the first equation also. Hence,  $a_1(b_2 c_3 - b_3 c_2) + b_1(c_2 a_3 - c_3 a_2) + c_1(a_2 b_3 - a_3 b_2) = 0$ .

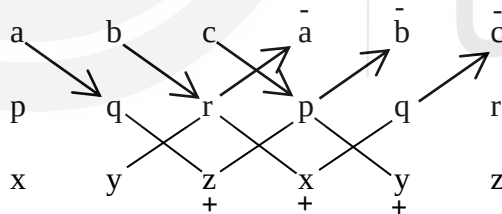
The expression  $a_1 b_2 c_3 - a_1 b_3 c_2 + a_3 b_1 c_2 - a_2 b_1 c_3 + a_2 b_3 c_1 - a_3 b_2 c_1$  is denoted by the symbol

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

and has three rows and three columns.

Therefore, it is called a '**determinant of the third order**' and it has  $3!=6$  terms; and to three out of these we prefix '+' sign and to the remaining three we prefix negative sign according to the following rule.

Rule for determining signs of involved terms in the expansion of  $3 \times 3$  determinant is given by the following figure in which the three columns of the determinant are written twice.



$$|A| \text{ (or) } \det A = aqz + brx + cpy - ary - bpz - cqx$$

In a 'determinant of  $n^{\text{th}}$  order', i.e., a determinant having n rows and n columns- there are  $n!$  terms in its expansion and out of them half of the terms would be affixed with positive signs and rest would be affixed with negative signs. According to the above rule, and as shown by the following expansion of a  $3 \times 3$  determinant:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2 c_3 - b_3 c_2) - b_1(a_2 c_3 - a_3 c_2) + c_1(a_2 b_3 - a_3 b_2)$$

### Determinant of a Square Matrix of Order 1

If  $A = [a_{11}]$  be a square matrix of order 1 then determinant of A is given by

$$|A| = |a_{11}| = a_{11}.$$

For example,

(i) If  $A = [5]$  then  $|A| = |5| = 5$ .

(ii) If  $A = [-3]$  then  $|A| = |-3| = -3$ .

#### Remark 1:

(i)  $|A|$  is read as determinant of A, do not read it modulus of A, i.e.

if  $A = [-8]$  then  $|A| = |-8| = -8$ .

But in case of modulus  $|-8| = -(-8) = 8$ .

(ii) The context in which we are using  $| \cdot |$  will clear whether it represents modulus or determinant.

### Determinant of a Square Matrix of Order $2 \times 2$

If  $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$  then  $|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$

Let us take an example:

**Example 1:** Evaluate the following determinants:

(i)  $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$  (ii)  $\begin{vmatrix} 3 & 5 \\ 8 & 9 \end{vmatrix}$  (iii)  $\begin{vmatrix} x^2 & x^2 + 1 \\ x & x + 1 \end{vmatrix}$

**Solution:**

(i)  $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

(ii)  $\begin{vmatrix} 3 & 5 \\ 8 & 9 \end{vmatrix} = 27 - 40 = -13$

(iii)  $\begin{vmatrix} x^2 & x^2 + 1 \\ x & x + 1 \end{vmatrix} = x^3 + x^2 - (x^3 + x) = x^2 - x$

### Determinant of a Square Matrix of Order $3 \times 3$

Before evaluating, the determinant of order  $3 \times 3$ , let us define the minors and cofactors of a square matrix as follows:

### 7.2.2 Minors and Cofactors

It may be noted that, in the last part of previous subsection, while evaluating a 3 x 3 determinant in terms of elements of first row,  $a_1$  is multiplied by a lower order determinant (of order 2). The second-order determinant  $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$  is obtained by deleting the column and row containing  $a_1$ . The determinant  $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$  is termed as ‘minor’ of the term  $a_1$  in the original matrix.

Same procedure is applied to obtain minors of other term also. Thus, the minor of  $b_2$  is obtained by deleting the 2<sup>nd</sup> row and 2<sup>nd</sup> column.

$$\text{In } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ minor of } b_2 \text{ is } \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$$

So, the minor of any element in an  $n^{\text{th}}$  order determinant is an  $(n-1)^{\text{th}}$  order determinant where  $n=1, 2, 3, \dots$  (any finite positive number).

#### Minor:

If  $A = [a_{ij}]_{n \times n}$  be a square matrix of order  $n$  then **minor of  $(i, j)^{\text{th}}$  element  $a_{ij}$  is denoted by  $M_{ij}$  and is defined as :**

$M_{ij}$  = determinant of the sub matrix of order  $n - 1$  obtained after deleting  $i^{\text{th}}$  row and  $j^{\text{th}}$  column from  $A$ .

**Example 2:** Find the minor of each element of the following matrices:

$$(i) \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix} \quad (ii) \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

#### Solution:

$$(i) \text{ Let } A = \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix}$$

Let  $M_{ij}$  denotes the minor of  $(i, j)^{\text{th}}$  element of the matrix  $A$ ,  $i, j = 1, 2$ .

$$\therefore M_{11} = |-7| = -7 \quad \left[ \begin{array}{l} \text{Determinant obtained after deleting first} \\ \text{row and first column of matrix } A = |-7| \end{array} \right]$$

$$\text{Similarly, } M_{12} = |4| = 4, \quad M_{21} = |5| = 5, \quad M_{22} = |2| = 2$$

$$(ii) \text{ Let } A = \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

Let  $M_{ij}$  denotes the minor of  $(i, j)^{th}$  element of the matrix A, where

$i, j = 1, 2, 3$ .

$$\therefore M_{11} = \begin{vmatrix} 5 & 7 \\ 9 & 1 \end{vmatrix} = 5 - 63 = -58 \quad \left[ \text{After deleting the first row and first column from A.} \right]$$

$$M_{12} = \begin{vmatrix} 6 & 7 \\ -8 & 1 \end{vmatrix} = 6 + 56 = 62 \quad \left[ \text{After deleting the first row and second column from A.} \right]$$

Similarly,

$$M_{13} = \begin{vmatrix} 6 & 5 \\ -8 & 9 \end{vmatrix} = 54 + 40 = 94$$

$$M_{21} = \begin{vmatrix} 4 & -2 \\ 9 & 1 \end{vmatrix} = 4 + 18 = 22$$

$$M_{22} = \begin{vmatrix} -3 & -2 \\ -8 & 1 \end{vmatrix} = -3 - 16 = -19$$

$$M_{23} = \begin{vmatrix} -3 & 4 \\ -8 & 9 \end{vmatrix} = -27 + 32 = 5$$

$$M_{31} = \begin{vmatrix} 4 & -2 \\ 5 & 7 \end{vmatrix} = 28 + 10 = 38$$

$$M_{32} = \begin{vmatrix} -3 & -2 \\ 6 & 7 \end{vmatrix} = -21 + 12 = -9$$

$$M_{33} = \begin{vmatrix} -3 & 4 \\ 6 & 5 \end{vmatrix} = -15 - 24 = -39$$

### Cofactor:

The 'cofactor' of any element in a determinant is its coefficient in the expansion of the determinant. It is therefore equal to the corresponding minor with a proper sign. The sign of the cofactor of an element in the  $i^{th}$  row and  $j^{th}$  column is  $(-1)^{i+j}$ .

Now,  $a_1$  lies in the 1<sup>st</sup> row and 1<sup>st</sup> column, hence its cofactor is

$$(-1)^{1+1} \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}.$$

Similarly,  $b_3$  lies in 3<sup>rd</sup> row and 2<sup>nd</sup> column,

$$\text{Hence, its cofactor is } (-1)^{3+2} \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}.$$

$$\text{Thus, } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 A_1 + b_1 B_1 + c_1 C_1 = a_1 A_1 + a_2 A_2 + a_3 A_3$$

where  $A_i, B_i, C_i$  ( $i=1, 2, 3$ ) are respective cofactors of  $a_i, b_i$  and  $c_i$ .

If  $A = [a_{ij}]_{n \times n}$  be a square matrix of order  $n$  then cofactor of  $(i, j)^{th}$  element

$a_{ij}$  of matrix  $A$  is denoted by  $C_{ij}$  and is defined by  $C_{ij} = (-1)^{i+j} M_{ij}$ , where

$M_{ij}$  denotes the minor of  $(i, j)^{th}$  element of the matrix  $A$ .

**Example 3:** Find the cofactors and the value of the following determinant

$$\begin{vmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ 7 & 2 & 1 \end{vmatrix}$$

**Solution:**

Here,  $a_1 = 3, b_1 = 4, c_1 = 7, a_2 = 2, b_2 = 1, c_2 = 3, a_3 = 7, b_3 = 2, c_3 = 1$

Here the cofactors are

$$A_1 = (-1)^{1+1} \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}$$

$$A_2 = (-1)^{1+2} \begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} = - \begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix}$$

$$A_3 = (-1)^{1+3} \begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix}$$

and so on.

So, expanding the determinant by 1<sup>st</sup> row, we have its value

$$= 3 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 4 \begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} + 7 \begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix}$$

$$= 3(1 - 6) - 4(2 - 21) + 7(4 - 7)$$

$$= -15 + 76 - 21 = 40$$

Similarly, we can expand the determinant by any row or column. If we expand it by the 1<sup>st</sup> column, then, value of the determinant will be

$$\begin{aligned}
 &= 3 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 7 \\ 2 & 1 \end{vmatrix} + 7 \begin{vmatrix} 4 & 7 \\ 1 & 3 \end{vmatrix} \\
 &= 3(1 - 6) - 2(4 - 14) + 7(12 - 7) \\
 &= -15 + 20 - 35 = 40
 \end{aligned}$$

As we are familiar with factors and cofactors, now, let us discuss the determinant of a square matrix of order  $3 \times 3$ .

If  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$  then

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Let us expand along first row( $R_1$ ), we have

$$\begin{aligned}
 |A| &= a_{11} (\text{co-factor of } a_{11}) + a_{12} (\text{co-factor of } a_{12}) + a_{13} (\text{co-factor of } a_{13}) \\
 &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\
 &= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})
 \end{aligned}$$

**Remark 2:**

- (i) We can expand the determinant along any row or column, we will get the same value.
- (ii) When we expand a determinant along any row or column we attach + or – sign with each term containing the product of elements of a row (or column) and its corresponding minor. Pattern of +, – signs for a  $3 \times 3$  determinant is shown as under.

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

We put + at (1, 1) position and then alternatively– and + are placed, provided either we can move along row or column (we cannot walk diagonally).

- (iii) There is no hard and fast rule, to choose a row or column to expand a

determinant. But if we choose that row or column which contains maximum number of zero, it will reduce our calculation work to great extent.

**Example 4:** Evaluate the following determinants:

$$(i) \begin{vmatrix} 3 & 2 & -1 \\ 5 & 4 & 6 \\ -3 & 1 & 7 \end{vmatrix} \quad (ii) \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} \quad (iii) \begin{vmatrix} 3 & 1 & 2 \\ 0 & 9 & 6 \\ 0 & 5 & 4 \end{vmatrix}$$

**Solution:**

$$(i) \text{ Let } \Delta = \begin{vmatrix} 3 & 2 & -1 \\ 5 & 4 & 6 \\ -3 & 1 & 7 \end{vmatrix}$$

Expanding along  $R_1$  (first row)

$$\Delta = 3(28 - 6) - 2(35 + 18) - 1(5 + 12) = 66 - 106 - 17 = -57$$

$$(ii) \text{ Let } \Delta = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix}$$

Expanding along  $R_1$  (first row)

$$\Delta = 2(2 - 4) - 1(1 - 4) + 2(2 - 4) = -4 + 3 - 4 = -5$$

$$(iii) \text{ Let } \Delta = \begin{vmatrix} 3 & 1 & 2 \\ 0 & 9 & 6 \\ 0 & 5 & 4 \end{vmatrix}$$

Expanding along  $C_1$  (first column)  $\left[ \because \text{it contains maximum number of zeros.} \right]$

$$\Delta = 3(36 - 30) - 0 + 0 = 18$$

**Remark 3:**

- (i) If A is square matrix then determinant of A is unique.
- (ii) If A is not a square matrix then determinant of A does not exist.

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### Check Your Progress - 1

1) Find x in each of the following cases:

$$(i) \begin{vmatrix} x & 7 \\ 9 & x+2 \end{vmatrix} = 0$$

(ii)  $\begin{vmatrix} x & x^2 \\ 15 & 5 \end{vmatrix} = 0$

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2) Find the cofactor of each element of the following matrices:

(i)  $\begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix}$

(ii)  $\begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$

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3) Find minor and cofactor of the elements  $a_{12}, a_{23}, a_{31}, a_{13}$  where

$$A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} 5 & 6 & -2 \\ -3 & 8 & 9 \\ 7 & 10 & -4 \end{bmatrix}$$

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4) If  $A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 5 & 1 \\ 2 & 4 & 8 \end{bmatrix}$  then show that  $|A| = 0$ .

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### 7.3 PROPERTIES OF DETERMINANTS

In Sec. 7.2 of this unit, you have become familiar about how to expand the determinants of orders 1, 2, 3, or of higher order. But as you have seen that it requires lot of calculations and is a time-consuming process. To avoid such calculations and to reduce the time of evaluation, we will use properties of determinants.

In this section, we will discuss some properties of the determinants. We shall give the proofs of some of these properties only for determinants of order  $3 \times 3$ . But remember that these properties hold good for all orders of the determinants. Let us see these properties one by one.

**Property → I** A determinant is unchanged in value, if all the rows of the determinant are changed into columns or vice versa, i.e.,

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{vmatrix}$$

**Example:**  $\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 3 \end{vmatrix}$  [following property (1)]

**Property → II** The value of a determinant is unaltered numerically but changed in sign if two rows (or two columns) are interchanged (sign of determinant is multiplied by  $(-1)$ ).

**Proof:** Let  $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$  ... (1)

Expanding along  $R_1$

$$\Delta = a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (2)$$

Let us interchange the first and second rows of the given determinant we have a new determinant  $\Delta_1$  (say) as:

$$\Delta_1 = \begin{vmatrix} x & y & z \\ a & b & c \\ l & m & n \end{vmatrix}$$

Expanding along  $R_1$

$$\begin{aligned} \Delta_1 &= x(bn - cm) - y(an - cl) + z(am - bl) \\ &= bnx - cmx - any + cly + amz - blz \\ &= -a(ny - mz) + b(nx - lz) - c(mx - ly) \\ &= -[a(ny - mz) - b(nx - lz) + c(mx - ly)] \\ &= -\Delta \quad [\text{Using (2)}] \end{aligned}$$

**Remark 4:** Here we interchanged  $R_1$  and  $R_2$ . In fact we can interchange any two rows or any two columns, result remains the same in each case.

**Example 5:**  $\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = - \begin{vmatrix} 3 & 1 & 0 \\ 4 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix}$  [by interchanging column (1)

and (2)]

$$= \begin{vmatrix} 4 & 2 & 0 \\ 3 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix} \quad [\text{by interchanging row (1) and (2)}]$$

**Property → III** A determinant is equal to zero if the elements of one row (or of one column) are equal or proportional to the corresponding elements of a second row (or of a second column). Or if any two rows or columns of a determinant are identical then value of the determinant vanishes.

**Proof:** Let  $\Delta = \begin{vmatrix} a & b & c \\ a & b & c \\ x & y & z \end{vmatrix}$ , where  $R_1$  and  $R_2$  are identical

Expanding along  $R_1$ , we get

$$\begin{aligned} \Delta &= a(bz - cy) - b(az - cx) + c(ay - bx) \\ &= abz - acy - abz + bcx + acy - bcx = 0 \end{aligned}$$

**Example 6:**  $\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 7 \\ 1 & 3 & 0 \end{vmatrix} = 0$  [since row (1) and (3) are identical]

**Example 7:**  $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 1 & 2 & 4 \end{vmatrix} = 0$  [since column (1) and (2) are proportional]

**Property → IV** If all the elements of any one row (or column) are multiplied (or divided) by K, then value of the determinant is also multiplied (or divided) by K, for division K is taken as non-zero.

$$\begin{vmatrix} Ka_{11} & Ka_{12} & \cdots & Ka_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = K \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

**Also,**

$$\begin{vmatrix} k_1 a_1 & k_1 b_1 & k_1 c_1 \\ k_2 a_2 & k_2 b_2 & k_2 c_2 \\ k_3 a_3 & k_3 b_3 & k_3 c_3 \end{vmatrix} = k_1 k_2 k_3 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

**Proof:** Let  $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$

Expanding along  $R_1$

$$\Delta = a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (1)$$

Let  $\Delta_1 = \begin{vmatrix} ka & b & c \\ kx & y & z \\ kl & m & n \end{vmatrix}$  [Here, the elements of first column of  $\Delta$  have been multiplied with k.]

Expanding along  $R_1$

$$\begin{aligned} \Delta_1 &= ka(ny - mz) - b(knx - klz) + c(kmx - kly) \\ &= k[a(ny - mz) - b(nx - lz) + c(mx - ly)] = k\Delta \quad \dots (2) \end{aligned}$$

[Using (1)]

From (1) and (2)

$$\Delta_1 = k\Delta$$

**Remark 5:** This property implies that if there is some factor common in all elements of any line then we can write it as the factor of the whole determinant.

**For example,**  $\begin{vmatrix} 5a & b & c \\ 5x & y & z \\ 5l & m & n \end{vmatrix} = 5 \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$

**Example 8:**

It can also be numerically shown by considering a 3<sup>rd</sup> order determinant

$$\begin{vmatrix} 2 & 10 & 16 \\ 3 & 7 & 1 \\ 5 & 4 & 11 \end{vmatrix} = 2 \begin{vmatrix} 1 & 5 & 8 \\ 3 & 7 & 1 \\ 5 & 4 & 11 \end{vmatrix} \quad [\text{since all the elements in the first row have a common factor 2}]$$

**Property** → **V** If all the elements of any row (or column) of a determinant can be expressed as the sum of (or difference between) two numbers, then the determinant can be expressed as the sum of (or difference between) two determinants.

**Example 9:**

$$\begin{vmatrix} a_1 + \alpha_1 & b_1 + \beta_1 & c_1 + \delta_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Numerically it can be shown as follow

$$\begin{vmatrix} 8+2 & 2+5 & 5+4 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix} = \Delta \text{ (say)} = \begin{vmatrix} 8 & 2 & 5 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix} + \begin{vmatrix} 2 & 5 & 4 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix}$$

$$= \Delta_1 + \Delta_2 \text{ (say)}$$

$$\therefore \Delta_1 = 8(63-11)-2(28-5)+5(44-45)$$

$$= 416-46-5=365$$

$$\therefore \Delta_2 = 2(63-11)-5(28-5)+4(44-45) = 104-115-4 = -15$$

$$\therefore \Delta_1 + \Delta_2 = 365 + (-15) = 365 - 15 = 350$$

$$\text{and } \Delta = 10(63-11)-7(28-5)+9(44-45) = 520-161-9=350$$

$$\therefore \Delta = \Delta_1 + \Delta_2$$

**Property** → **VI** The value of a determinant remains unchanged by adding (or subtracting) k times the elements of any row (or column) to (or from) the corresponding elements of any other row (or column), where k is any given number.

**Example 10 :**

$$\begin{vmatrix} a_1 + kb_1 & b_1 & c_1 \\ a_2 + kb_2 & b_2 & c_2 \\ a_3 + kb_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} kb_1 & b_1 & c_1 \\ kb_2 & b_2 & c_2 \\ kb_3 & b_3 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + k \begin{vmatrix} b_1 & b_1 & c_1 \\ b_2 & b_2 & c_2 \\ b_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

[since the value of the second determinant is zero as its 1<sup>st</sup> and 2<sup>nd</sup> columns are identical]

**Property → VII**

$|A'| = |A|$ , i.e. determinants of a matrix and its transpose are equal.

**Proof:** Let

$$A = \begin{bmatrix} a & b & c \\ x & y & z \\ l & m & n \end{bmatrix} \quad \dots (1)$$

$$|A| = \begin{vmatrix} a & b & c \\ x & y & z \\ l & m & n \end{vmatrix}$$

Expanding along  $R_1$

$$|A| = a(ny - mz) - b(nx - lz) + c(mx - ly) \quad \dots (2)$$

From (1), we get

$$A' = \begin{bmatrix} a & x & l \\ b & y & m \\ c & z & n \end{bmatrix}$$

$$\therefore |A'| = \begin{vmatrix} a & x & l \\ b & y & m \\ c & z & n \end{vmatrix}$$

Expanding along  $R_1$

$$\begin{aligned} |A'| &= a(ny - mz) - x(bn - cm) + l(bz - cy) \\ &= a(ny - mz) - bnx + cmx + lbz - cly \\ &= a(ny - mz) - b(nx - lz) + c(mx - ly) \end{aligned} \quad \dots (3)$$

From (2) and (3), we get

$$|A'| = |A|$$

**Property → VIII**

If all the elements of any line (row or column) are zero then value of the determinant vanishes.

**Proof:** Let  $\Delta = \begin{vmatrix} 0 & 0 & 0 \\ x & y & z \\ l & m & n \end{vmatrix}$ , then evaluating along  $R_1$ , we get

$$\Delta = (0)(ny - mz) - (0)(nx - lz) + (0)(mx - ly) = 0 - 0 + 0 = 0$$

**Example 11:** Evaluate the following determinants:

$$(i) \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} \quad (ii) \begin{vmatrix} -3 & 5 & -2 \\ 8 & 9 & -17 \\ 3 & -6 & 3 \end{vmatrix}$$

$$(iii) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

$$(iv) \begin{vmatrix} 3 & x & xyz \\ 3 & y & xyz \\ 3 & z & xyz \end{vmatrix}$$

$$(v) \begin{vmatrix} 2 & 3 & 30 \\ 5 & 4 & 54 \\ 6 & 1 & 42 \end{vmatrix}$$

$$(vi) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$(vii) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$(viii) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix}$$

$$(ix) \begin{vmatrix} 2x+3 & x & x \\ x & 2x+3 & x \\ x & x & 2x+3 \end{vmatrix}$$

**Solution:**

$$(i) \text{ Let } \Delta = \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

Operating  $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} 0 & b-c & c-a \\ 0 & c-a & a-b \\ 0 & a-b & b-c \end{vmatrix} = 0 \quad [\because \text{all elements of } C_1 \text{ are zero, so using P7.}]$$

$$(ii) \text{ Let } \Delta = \begin{vmatrix} -3 & 5 & -2 \\ 8 & 9 & -17 \\ 3 & -6 & 3 \end{vmatrix}$$

Operating  $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} -3+5-2 & 5 & -2 \\ 8+9-17 & 9 & -17 \\ 3-6+3 & -6 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 5 & -2 \\ 0 & 9 & -17 \\ 0 & -6 & 3 \end{vmatrix} = 0 \quad [\because \text{all the element of } C_1 \text{ are zero, so using P7.}]$$

$$(iii) \text{ Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

Operating  $R_3 \rightarrow R_3 + R_2$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a+b+c & a+b+c & a+b+c \end{vmatrix}$$

Taking  $(a + b + c)$  common from  $R_3$

$$\Delta = (a + b + c) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = (a + b + c)(0) = 0 [\because R_1 \text{ and } R_3 \text{ are identical.}]$$

$$(iv) \text{ Let } \Delta = \begin{vmatrix} 3 & x & xyz \\ 3 & y & xyz \\ 3 & z & xyz \end{vmatrix}$$

Taking 3, xyz common from  $C_1$  and  $C_3$  respectively

$$\Delta = 3xyz \begin{vmatrix} 1 & x & 1 \\ 1 & y & 1 \\ 1 & z & 1 \end{vmatrix} = 3xyz (0) = 0 \quad [\because C_1 \text{ and } C_3 \text{ are identical.}]$$

$$(v) \text{ Let } \Delta = \begin{vmatrix} 2 & 3 & 30 \\ 5 & 4 & 54 \\ 6 & 1 & 42 \end{vmatrix}$$

Taking 6 common from  $C_3$

$$\Delta = 6 \begin{vmatrix} 2 & 3 & 5 \\ 5 & 4 & 9 \\ 6 & 1 & 7 \end{vmatrix}$$

Operating  $C_3 \rightarrow C_3 - C_1 - C_2$

$$\Delta = 6 \begin{vmatrix} 2 & 3 & 0 \\ 5 & 4 & 0 \\ 6 & 1 & 0 \end{vmatrix} = 6(0) = 0$$

[ $\because$  all the elements of  $C_3$  are zero, so using P7.]

$$(vi) \text{ Let } \Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

Operating  $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 0 & (y-x) & (y-x).1 & (y-x)(y+x) \\ 0 & (z-x) & (z-x).1 & (z-x)(z+x) \end{vmatrix}$$

Taking  $y-x, z-x$  common from  $R_2, R_3$  respectively

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 1 & z+x \end{vmatrix}$$

Operating  $R_3 \rightarrow R_3 - R_2$

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & z-y \end{vmatrix}$$

$$= (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0.(z-y) & 0.(z-y) & 1.(z-y) \end{vmatrix}$$

Taking  $(z-y)$  common from  $R_3$

$$\Delta = (y-x)(z-x)(z-y) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along  $C_1$ , we get :

$$\Delta = (y-x)(z-x)(z-y)[1(1-0)-0+0]$$

$$= (x-y)(y-z)(z-x)$$

(vii) Let  $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Operating  $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} a+b+c & b & c \\ a+b+c & c & a \\ a+b+c & a & b \end{vmatrix}$$

Taking  $(a+b+c)$  common from  $C_1$

$$\Delta = (a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

Operating  $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\Delta = (a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix}$$

Expanding along  $C_1$

$$\Delta = (a+b+c)[1\{(c-b)(b-c)-(a-b)(a-c)\}-0+0]$$

$$= (a+b+c)[bc-c^2-b^2+bc-(a^2-ac-ab+bc)]$$

$$= (a+b+c)(ab+bc+ca-a^2-b^2-c^2)$$

$$= 3abc-a^3-b^3-c^3$$

(viii) Let  $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix}$

Operating  $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^3 & y^3-x^3 & z^3-x^3 \end{vmatrix}$$

Taking  $y-x, z-x$  common from  $C_2, C_3$  respectively

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^3 & y^2 + x^2 + xy & z^2 + x^2 + zx \end{vmatrix}$$

Operating  $C_3 \rightarrow C_3 - C_2$

$$\Delta = (y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^3 & x^2 + y^2 + xy & z^2 - y^2 + x(z-y) \end{vmatrix}$$

$$= (y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^3 & x^2 + y^2 + xy & (z-y)(z+y+x) \end{vmatrix}$$

Taking  $(z-y)(x+y+z)$  common from  $C_3$

$$\Delta = (y-x)(z-x)(z-y)(x+y+z) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^3 & x^2 + y^2 + xy & 1 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta = (y-x)(z-x)(z-y)(x+y+z)[1(1-0)-0+0]$$

$$= (x-y)(y-z)(z-x)(x+y+z)$$

$$(ix) \text{ Let } \Delta = \begin{vmatrix} 2x+3 & x & x \\ x & 2x+3 & x \\ x & x & 2x+3 \end{vmatrix}$$

Operating  $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Delta = \begin{vmatrix} 4x+3 & x & x \\ 4x+3 & 2x+3 & x \\ 4x+3 & x & 2x+3 \end{vmatrix}$$

Taking  $4x+3$  common from  $C_1$

$$\Delta = (4x+3) \begin{vmatrix} 1 & x & x \\ 1 & 2x+3 & x \\ 1 & x & 2x+3 \end{vmatrix}$$

Operating  $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\Delta = (4x+3) \begin{vmatrix} 1 & x & x \\ 0 & x+3 & 0 \\ 0 & 0 & x+3 \end{vmatrix}$$

Expanding along  $C_1$

$$\Delta = (4x+3)[1\{(x+3)^2 - 0\} - 0 + 0] = (4x+3)(x+3)^2$$

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### Check Your Progress -2

1) Prove the following

$$\begin{vmatrix} ab & 1 & c(a+b) \\ bc & 1 & a(b+c) \\ ca & 1 & b(c+a) \end{vmatrix} = 0 \quad \text{[Without expanding]}$$

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2) Prove the following

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1+x^3 & 1+y^3 & 1+z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(1+xyz)$$

[Using properties]

3) Prove the following

$$\begin{vmatrix} a+b & c & c \\ a & b+c & a \\ b & b & c+a \end{vmatrix} = 4abc \quad \text{[Using properties]}$$

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## 7.4 PRODUCT OF DETERMINANTS

It is noteworthy that determinants can be multiplied together only if they are of the same order. As the process of interchanging the rows and columns will not affect the value of the determinant (by Property- I). Hence, we can also adopt the following procedures for multiplication of two determinants.

- (i) Row by column multiplication rule
- (ii) Row by row multiplication rule
- (iii) Column by column multiplication rule
- (iv) Column by row multiplication rule

### Product of Two Determinants

The product of two determinants of same order is another determinant of the same order.

The product of two determinants (same order, e.g.,  $n \times n$ ) results in a new determinant, calculated by performing matrix multiplication (row-by-column, column-by-column, etc.) on their corresponding matrices, with the key property that  $\det(A * B) = \det(A) * \det(B)$ . This means you can find the determinant of the product matrix or simply multiply the individual determinant values to get the same scalar result.

In the row by column rule of Determinants multiplication, every element of each row of a determinant is multiplied by every element of every column of another determinant. Remember, this rule is the same as when two matrices are multiplied.

**Example 12:**  $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} c_1 & d_1 \\ c_2 & d_2 \end{vmatrix} = \begin{vmatrix} a_1c_1 + b_1d_1 & a_1c_2 + b_1d_2 \\ a_2c_1 + b_2d_1 & a_2c_2 + b_2d_2 \end{vmatrix}$

**Example 13:**  $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} =$

Let us see an example,  $\begin{vmatrix} 4 & 5 & 9 \\ 1 & 2 & 3 \\ 3 & 0 & 1 \end{vmatrix} \times \begin{vmatrix} 2 & 1 & 5 \\ 3 & 2 & 1 \\ 1 & 6 & 4 \end{vmatrix}$

$$= \begin{vmatrix} 4 \times 2 + 5 \times 3 + 9 \times 1 & 4 \times 1 + 5 \times 2 + 9 \times 6 & 4 \times 5 + 5 \times 1 + 9 \times 4 \\ 1 \times 2 + 2 \times 3 + 3 \times 1 & 1 \times 1 + 2 \times 2 + 3 \times 6 & 1 \times 5 + 2 \times 1 + 3 \times 4 \\ 3 \times 2 + 0 \times 3 + 2 \times 1 & 3 \times 1 + 0 \times 2 + 2 \times 6 & 3 \times 5 + 0 \times 1 + 2 \times 4 \end{vmatrix}$$

$$= \begin{vmatrix} 32 & 68 & 61 \\ 11 & 23 & 19 \\ 8 & 15 & 23 \end{vmatrix}$$

## 7.5 ADJOINT AND RECIPROCAL DETERMINANTS

The '*adjoint*' or '*adjugate*' of a determinant  $\Delta$  is the determinant  $\Delta'$  whose elements are the cofactors of the corresponding elements of  $\Delta$

Let  $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  then,  $\Delta^T = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$

where  $A_1, B_1, C_1, \dots$  are the respective cofactors of  $a_1, b_1, c_1, \dots$  of  $\Delta$ .

**Note:** Here  $\Delta^T = \Delta^2$

**Example 14:** Let  $\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{vmatrix}$

Here,  $a_1 = 0, b_1 = 1, c_1 = 2$

$a_2 = 2, b_2 = 0, c_2 = 1$

$a_3 = 1, b_3 = 2, c_3 = 0$

$$\therefore A_1 = \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = 0 - 2 = -2$$

$$B_1 = (-1) \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -(0 - 1) = +1$$

Similarly,  $C_1 = 4$ ,  $A_2 = 4$ ,  $B_2 = -2$ ,  $C_2 = 1$ ,

$A_3 = 1$ ,  $B_3 = 4$ ,  $C_3 = -2$

$$\therefore \text{Adjoint of } \Delta \text{ is } \Delta' = \begin{vmatrix} -2 & 1 & 4 \\ 4 & -2 & 1 \\ 1 & 4 & -2 \end{vmatrix}$$

The ‘**reciprocal**’ or ‘**inverse**’ of a determinant  $\Delta$  ( $\neq 0$ ) is the determinant, which is formed by dividing every element of the adjoint of  $\Delta$  by  $\Delta$ . Thus, the reciprocal of  $\Delta$  is

$$\begin{vmatrix} \frac{A_1}{\Delta} & \frac{B_1}{\Delta} & \frac{C_1}{\Delta} \\ \frac{A_2}{\Delta} & \frac{B_2}{\Delta} & \frac{C_2}{\Delta} \\ \frac{A_3}{\Delta} & \frac{B_3}{\Delta} & \frac{C_3}{\Delta} \end{vmatrix} = \frac{1}{\Delta^3} \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = \frac{1}{\Delta^3} \cdot \Delta' = \frac{1}{\Delta^3} \cdot \Delta^2 = \Delta^{-1}$$

**Example 15:** Following the last example where  $\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{vmatrix}$ , we get,

$$\Delta = 0(0-2)+1(1-0)+2(4-0) = 9 \neq 0$$

$$\therefore \text{The reciprocal of } \Delta \text{ is: } \begin{vmatrix} -\frac{2}{9} & \frac{1}{9} & \frac{4}{9} \\ 4/9 & -2/9 & 1/9 \\ 1/9 & 4/9 & -2/9 \end{vmatrix}$$

## 7.6 SYMMETRIC DETERMINANTS

If a 3 x 3 determinant is written in the form  $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

in which the suffix of any element indicates its position (e.g., the suffix 12 of the element  $a_{12}$  indicates that the element is in the 1<sup>st</sup> row and in the 2<sup>nd</sup> column) and if  $a_{ij} = a_{ji}$  for all  $i, j = 1, 2, 3$ ; then the **determinant is said to be symmetric**.

Thus  $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$  is a symmetric determinant.

The above definition of symmetric determinant can be extended similarly to determinants of order  $n$ .

- Properties:**
- (i) The square of any symmetric determinant is a symmetric determinant.
  - (ii) The adjoint of symmetric determinant is symmetric.

**Example 16:** Let  $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix}$

Here  $\Delta$  is a symmetric determinant

$$\begin{aligned} \text{Now, } \Delta^2 &= \Delta \cdot \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix} \times \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix} \\ &= \begin{vmatrix} 1+4+9 & 2+8+15 & 3+10+21 \\ 2+8+15 & 4+16+25 & 6+20+35 \\ 3+10+21 & 6+20+35 & 9+25+49 \end{vmatrix} \\ &= \begin{vmatrix} 14 & 25 & 34 \\ 25 & 45 & 61 \\ 34 & 61 & 83 \end{vmatrix}, \text{ which is also a symmetric determinant.} \end{aligned}$$

$$\text{Now adj. } \Delta = \begin{vmatrix} 3 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 0 \end{vmatrix}, \text{ which is also a symmetric determinant}$$

### 7.6.1 Skew and Skew-Symmetric Determinants

A skew determinant refers to the determinant of a matrix that has specific symmetry properties.

The determinant  $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$  is said to be 'skew', if  $a_{ij} = -a_{ji} \forall i, j = 1, 2, 3$  and  $i \neq j$

Thus,  $\begin{vmatrix} a & -h & -g \\ h & b & -f \\ g & f & c \end{vmatrix}$  is a **skew determinant**.

and if  $a_{ij} = -a_{ji} \forall i, j = 1, 2, 3$  and  $i \neq j$  with  $a_{ii} = 0 \forall i = j$ ,

then the determinant is said to be **skew-symmetric**.

Thus,  $\begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix}$  is a **skew-symmetric** determinant.

**Properties:** (i) Every skew-symmetric determinant of third order (more generally, of odd order) is zero.,

(ii) Every skew-symmetric determinant of second order (more generally of even order) is a perfect square.

**Example 17:** Let  $\Delta = \begin{vmatrix} 0 & 5 & 2 \\ -5 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix} = (-1)^3 \begin{vmatrix} 0 & -5 & -2 \\ 5 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix} = -\Delta$

[taking (-1) common from each column]

Hence,  $2\Delta = 0 \Rightarrow \Delta = 0$

Again, let us consider an even order determinant

$$\Delta_1 = \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix}$$

$\therefore \Delta_1 = 0 - (-4) = 4 = 2^2$  which is a perfect square.

## 7.7 APPLICATION OF DETERMINANTS: SOLUTION OF SIMULTANEOUS EQUATIONS BY CRAMER'S RULE

There are many applications of determinants but here we shall discuss as to how the concept of determinant provides the solution of a given system of linear equations. The procedure which we will use is known as Cramer's rule.

**Cramer's Rule:** Cramer's rule can be used in any system of **n linear equations in n variables**. But here we shall discuss Cramer's rule only for system of 2 (or 3) equations in 2 (or 3) variables respectively. **Gabriel Cramer** obtained a simple method of solution of linear simultaneous equations by using the properties of determinants.

In particular, let the system of 3 equations in 3 variables, be

$$a_i x + b_i y + c_i z = k_i \quad (i=1,2,3) \dots\dots\dots(1)$$

$$\text{and } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$$

$$\text{Then } x\Delta = \begin{vmatrix} xa_1 & b_1 & c_1 \\ xa_2 & b_2 & c_2 \\ xa_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1x + b_1y + c_1z & b_1 & c_1 \\ a_2x + b_2y + c_2z & b_2 & c_2 \\ a_3x + b_3y + c_3z & b_3 & c_3 \end{vmatrix}$$

$$[c_i' = c_1 + yc_2 + zc_3]$$

$$= \begin{vmatrix} k_1 & b_1 & c_1 \\ k_2 & b_2 & c_2 \\ k_3 & b_3 & c_3 \end{vmatrix} \quad [\text{from (1)}] = \Delta_1 \text{ (say)}$$

$$\therefore \Delta = \frac{\Delta_1}{\Delta}$$

Similarly,  $y = \frac{\Delta_2}{\Delta}$  and  $z = \frac{\Delta_3}{\Delta}$  where  $\Delta_2$  and  $\Delta_3$  are the determinants obtained from  $\Delta$  substituting the columns  $(k_1, k_2, k_3)$  for the second the third column respectively.

**Example 18:** Suppose that the numbers  $x_1$  and  $x_2$  satisfy the equations

$$x_1 - 2x_2 = 3 \text{ and } 3x_1 + 5x_2 = 20.$$

Find  $x_1$  and  $x_2$  by using Cramer's rule.

**Solution:**

$$x_1 - 2x_2 = 3,$$

$$3x_1 + 5x_2 = 20$$

It may be written as matrix equation  $AX = B$

$$\text{where, } A = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 \\ 20 \end{pmatrix}$$

Cramer's rule tells us that

$$x_1 = \frac{\begin{vmatrix} 3 & -2 \\ 20 & 5 \end{vmatrix}}{\begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix}} = \frac{3(5) - (-2)(20)}{1(5) - (-2)3} = \frac{55}{11} = 5$$

$$\text{and } x_2 = \frac{\begin{vmatrix} 1 & 3 \\ 3 & 20 \end{vmatrix}}{\begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix}} = \frac{1(20) - 3(3)}{11} = \frac{11}{11} = 1$$

Thus, if we have the value of the original determinant along with the values obtained from substituting the column of constants (i.e.,  $k_i$ ) in different places, then by Cramer's rule we can have the value of the corresponding variable.

**Example 19 :** Solve the following system of equations using Cramer's rule:

$$2x + 3y + z = 4$$

$$x - y + 2z = 9$$

$$3x + 2y - z = 1$$

**Solution:** Given system of equations is

$$2x + 3y + z = 4 \quad \dots (1)$$

$$x - y + 2z = 9 \quad \dots (2)$$

$$3x + 2y - z = 1 \quad \dots (3)$$

$$\text{Here, } \Delta = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -1 & 2 \\ 3 & 2 & -1 \end{vmatrix}$$

Expanding along  $R_1$ , we get

$$\Delta = 2(1 - 4) - 3(-1 - 6) + 1(2 + 3) = -6 + 21 + 5 = 20 \neq 0$$

$\therefore$  system has unique solution.

$$\Delta_1 = \begin{vmatrix} 4 & 3 & 1 \\ 9 & -1 & 2 \\ 1 & 2 & -1 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta_1 = 4(1 - 4) - 3(-9 - 2) + 1(18 + 1) = -12 + 33 + 19 = 40$$

$$\Delta_2 = \begin{vmatrix} 2 & 4 & 1 \\ 1 & 9 & 2 \\ 3 & 1 & -1 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta_2 = 2(-9 - 2) - 4(-1 - 6) + 1(1 - 27) = -22 + 28 - 26 = -20$$

$$\Delta_3 = \begin{vmatrix} 2 & 3 & 4 \\ 1 & -1 & 9 \\ 3 & 2 & 1 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta_3 = 2(-1 - 18) - 3(1 - 27) + 4(2 + 3) = -38 + 78 + 20 = 60$$

$\therefore$  by Cramer's rule

$$x = \frac{\Delta_1}{\Delta} = \frac{40}{20} = 2, y = \frac{\Delta_2}{\Delta} = \frac{-20}{20} = -1, z = \frac{\Delta_3}{\Delta} = \frac{60}{20} = 3$$

$$\therefore x = 2, y = -1, z = 3$$

**Example 20 :** Solve the following system of equations:

$$x + 3y + 2z = 6$$

$$-x + 4y + 5z = 8$$

$$2x + 5y + 3z = 10$$

**Solution:** Given system of equations is

$$x + 3y + 2z = 6 \quad \dots (1)$$

$$-x + 4y + 5z = 8 \quad \dots (2)$$

$$2x + 5y + 3z = 10 \quad \dots (3)$$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & 3 & 2 \\ -1 & 4 & 5 \\ 2 & 5 & 3 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta = 1(12 - 25) - 3(-3 - 10) + 2(-5 - 8) = -13 + 39 - 26 = -39 + 39 = 0$$

$\therefore$  system has either no solution or infinite many solutions.

$$\text{Now, } \Delta_1 = \begin{vmatrix} 6 & 3 & 2 \\ 8 & 4 & 5 \\ 10 & 5 & 3 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta_1 = 6(12 - 25) - 3(24 - 50) + 2(40 - 40) = -78 + 78 + 0 = 0$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 2 \\ -1 & 8 & 5 \\ 2 & 10 & 3 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta_2 = 1(24 - 50) - 6(-3 - 10) + 2(-10 - 16) = -26 + 78 - 52 = -78 + 78 = 0$$

$$\Delta_3 = \begin{vmatrix} 1 & 3 & 6 \\ -1 & 4 & 8 \\ 2 & 5 & 10 \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta_3 = 1(40 - 40) - 3(-10 - 16) + 6(-5 - 8) = 0 + 78 - 78 = 0$$

As  $\Delta = 0$  and also  $\Delta_1 = \Delta_2 = \Delta_3 = 0$

$\Rightarrow$  system has infinite many solutions which are given by putting  $z = k$  (where  $k$  is any real number) in any of the two equations given by (1), (2) and (3).

Let us put  $z = k$  in (1) and (2), we get

$$x + 3y = 6 - 2k \quad \dots (4)$$

$$-x + 4y = 8 - 5k \quad \dots (5)$$

Again for this system of equations

$$\Delta = \begin{vmatrix} 1 & 3 \\ -1 & 4 \end{vmatrix} = 4 + 3 = 7$$

$$\Delta_1 = \begin{vmatrix} 6 - 2k & 3 \\ 8 - 5k & 4 \end{vmatrix} = 24 - 8k - 24 + 15k = 7k$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 - 2k \\ -1 & 8 - 5k \end{vmatrix} = 8 - 5k + 6 - 2k = 14 - 7k$$

$\therefore$  by Cramer's rule

$$x = \frac{\Delta_1}{\Delta} = \frac{7k}{7} = k, \quad y = \frac{\Delta_2}{\Delta} = \frac{14 - 7k}{7} = 2 - k$$

$\therefore x = k, y = 2 - k, z = k$ , where  $k$  is any real number

**Remark 6:** Here, in the above example, we have taken  $z = k$ . You may also take  $y = k$  or  $x = k$  and then can solve by taking any two equations out of (1), (2), (3) in remaining two unknowns using Cramer's rule.

Now, you can try the following exercises.

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### Check Your Progress – 3

1. Solve the following system of equations using Cramer's rule:

$$(i) \quad 3x + 5y = -11 \quad \dots (1)$$

$$2x - 3y = 18 \quad \dots (2)$$

$$(ii) \quad x - 2y = 5 \quad \dots (1)$$

$$-2x + 4y = 8 \quad \dots (2)$$

$$(iii) \quad 2x - y = 6 \quad \dots (1)$$

$$-6x + 3y = -18 \quad \dots (2)$$

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2. Solve the following system of equations:

$$x + y + 2z = 4$$

$$x - y + 3z = 3$$

$$2x + 2y + 4z = 7$$

using Cramer's rule.

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3. Solve the following system of equations by Cramer's rule

$$x + 2y + 3z = 6 \quad \dots (1)$$

$$2x + 4y + z = 7 \quad \dots (2)$$

$$3x + 2y + 9z = 14 \quad \dots (3)$$

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## 7.8 SUMMARY

This unit provides an exhaustive study of determinants, beginning with the foundational definitions and the essential mechanics of calculating minors and cofactors to evaluate higher-order values. This unit provides details the structural properties that allow for the simplification of complex determinants, such as row/column interchanges, scalar multiplication, and the zero-value rule for identical rows. The unit explains advanced operations, including the product of two determinants and the derivation of adjoint and reciprocal determinants, which are vital for matrix inversion. It further helps in distinguishing between symmetric determinants and the unique properties of skew and skew-symmetric forms, where the order of the determinant often dictates its final value. Finally, the theoretical concepts culminate in a practical application through Cramer's Rule, providing a systematic method for solving simultaneous linear equations by utilizing the ratios of specific coefficient determinants.

## 7.9 SOLUTIONS/ANSWER TO CHECK YOUR PROGRESS

### Check Your Progress - 1

1. (i)

$$\begin{vmatrix} x & 7 \\ 9 & x+2 \end{vmatrix} = 0$$

$$\Rightarrow x(x+2) - 63 = 0$$

$$\Rightarrow x^2 + 2x - 63 = 0 \Rightarrow x^2 + 9x - 7x - 63 = 0$$

$$\Rightarrow x(x+9) - 7(x+9) = 0$$

$$\Rightarrow (x+9)(x-7) = 0 \Rightarrow x = 7, -9$$

1. (ii)

$$\begin{vmatrix} x & x^2 \\ 15 & 5 \end{vmatrix} = 0$$

$$\Rightarrow 5x - 15x^2 = 0$$

$$\Rightarrow 15x^2 - 5x = 0 \Rightarrow 5x(3x-1) = 0 \Rightarrow x = 0, 1/3$$

2. (i)

$$\text{Let } A = \begin{bmatrix} 2 & 5 \\ 4 & -7 \end{bmatrix}$$

Let  $C_{ij}$  denotes the cofactor of  $(i, j)^{th}$  element of the matrix  $A$ ,  $i, j = 1, 2$ .

$$\therefore C_{11} = (-1)^{1+1}M_{11} = (-1)^2(-7) = -7 \quad [\text{Using Example 11 (i)}]$$

$$\text{Similarly, } C_{12} = (-1)^{1+2}M_{12} = (-1)^3(4) = -4$$

$$C_{21} = (-1)^{2+1}M_{21} = (-1)^3(5) = -5$$

$$C_{22} = (-1)^{2+2}M_{22} = (-1)^4(2) = 2$$

2. (ii)

$$\text{Let } A = \begin{bmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{bmatrix}$$

Let  $C_{ij}$  denotes the cofactor of  $(i, j)^{th}$  element of the matrix A, then

$$C_{11} = (-1)^{1+1}M_{11} = (-1)^2(-58) = -58 \quad [\text{Using Example 11 (ii)}]$$

Similarly,

$$C_{12} = (-1)^{1+2}M_{12} = (-1)^3(62) = -62$$

$$C_{13} = (-1)^{1+3}M_{13} = (-1)^4(94) = 94$$

$$C_{21} = (-1)^{2+1}M_{21} = (-1)^3(22) = -22$$

$$C_{22} = (-1)^{2+2}M_{22} = (-1)^4(-19) = -19$$

$$C_{23} = (-1)^{2+3}M_{23} = (-1)^5(5) = -5$$

$$C_{31} = (-1)^{3+1}M_{31} = (-1)^4(38) = 38$$

$$C_{32} = (-1)^{3+2}M_{32} = (-1)^5(-9) = 9$$

$$C_{33} = (-1)^{3+3}M_{33} = (-1)^6(-39) = -39$$

3.

Let  $M_{ij}$  and  $C_{ij}$  denote the minor and cofactor of  $(i, j)^{th}$  element in the matrix A respectively then

$$M_{12} = \begin{vmatrix} -3 & 9 \\ 7 & -4 \end{vmatrix} = 12 - 63 = -51 \quad M_{23} = \begin{vmatrix} 5 & 6 \\ 7 & 10 \end{vmatrix} = 50 - 42 = 8$$

$$M_{31} = \begin{vmatrix} 6 & -2 \\ 8 & 9 \end{vmatrix} = 54 - (-16) = 54 + 16 = 70$$

$$M_{13} = \begin{vmatrix} -3 & 8 \\ 7 & 10 \end{vmatrix} = -30 - 56 = -86$$

$$C_{12} = (-1)^{1+2}M_{12} = (-1)^3(-51) = 51, \quad C_{23} = (-1)^{2+3}M_{23} = (-1)^5(8) = -8$$

$$C_{31} = (-1)^{3+1}M_{31} = (-1)^4(70) = 70$$

$$C_{13} = (-1)^{1+3}M_{13} = (-1)^4(-86) = -86$$

4.

$$|A| = \begin{vmatrix} 1 & 2 & 4 \\ -3 & 5 & 1 \\ 2 & 4 & 8 \end{vmatrix}$$

Expanding along  $R_1$

$$|A| = 1(40 - 4) - 2(-24 - 2) + 4(-12 - 10) = 36 + 52 - 88 = 0$$

### Check Your Progress -2

1.

$$\text{L.H.S.} = \begin{vmatrix} ab & 1 & c(a+b) \\ bc & 1 & a(b+c) \\ ca & 1 & b(c+a) \end{vmatrix} = \begin{vmatrix} ab & 1 & ac+bc \\ bc & 1 & ab+ac \\ ca & 1 & bc+ab \end{vmatrix}$$

Operating  $C_3 \rightarrow C_3 + C_1$

$$\text{L.H.S.} = \begin{vmatrix} ab & 1 & ab+bc+ca \\ bc & 1 & ab+bc+ca \\ ca & 1 & ab+bc+ca \end{vmatrix}$$

Taking  $ab + bc + ca$  common from  $C_3$

$$\text{L.H.S.} = (ab + bc + ca) \begin{vmatrix} ab & 1 & 1 \\ bc & 1 & 1 \\ ca & 1 & 1 \end{vmatrix}$$

$$= (ab + bc + ca)(0) = 0 = \text{R.H.S.}$$

[ $\because C_2$  and  $C_3$  are identical.]

2.

$$\text{Let } \Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1+x^3 & 1+y^3 & 1+z^3 \end{vmatrix}$$

$$= \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix}$$

[Using property 5]

Taking  $x, y, z$  common from  $C_1, C_2, C_3$  of the second determinant respectively.

$$\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Operating  $R_1 \leftrightarrow R_3$  on first determinant

$$\Delta = (-1) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Operating  $R_2 \leftrightarrow R_3$  on first determinant

$$\Delta = (-1)(-1) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Taking determinant common from both terms

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Operating  $C_2 \rightarrow C_2 - C_1; C_3 \rightarrow C_3 - C_1$

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & 0 & 0 \\ x & y - x & z - x \\ x^2 & y^2 - x^2 & z^2 - x^2 \end{vmatrix}$$

Taking  $y - x, z - x$  common from  $C_2, C_3$  respectively

$$\Delta = (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^2 & x + y & z + x \end{vmatrix}$$

Operating  $C_3 \rightarrow C_3 - C_2$

$$\Delta = (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x^2 & x + y & z - y \end{vmatrix}$$

Expanding along  $R_1$

$$\Delta = (1 + xyz)(y - x)(z - x)[1\{(z - y) - 0\} - 0 + 0]$$

$$= (1 + xyz)(y - x)(z - x)(z - y) = (x - y)(y - z)(z - x)(1 + xyz) = \text{R.H.S.}$$

3.

$$\text{L.H.S} = \begin{vmatrix} a + b & c & c \\ a & b + c & a \\ b & b & c + a \end{vmatrix}$$

Operating  $R_1 \rightarrow R_1 - R_2 - R_3$

$$\text{L.H.S} = \begin{vmatrix} 0 & -2b & -2a \\ a & b + c & a \\ b & b & c + a \end{vmatrix}$$

Operating  $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$

$$\text{L.H.S} = \begin{vmatrix} 0 & -2b & -2a \\ a & b + c - a & 0 \\ b & 0 & c + a - b \end{vmatrix}$$

Expanding along  $R_1$

$$\text{L.H.S} = 0 - (-2b)[a(c + a - b) - 0] + (-2a)[0 - b(b + c - a)]$$

$$= 2ab(c + a - b) + 2ab(b + c - a) = 2ab(c + a - b + b + c - a)$$

$$= 2ab(2c) = 4abc = \text{R.H.S.}$$

### Check Your Progress -3

#### 1. (i)

Given system of equations is

$$3x + 5y = -11 \quad \dots (1)$$

$$2x - 3y = 18 \quad \dots (2)$$

$$\text{Here, } \Delta = \begin{vmatrix} 3 & 5 \\ 2 & -3 \end{vmatrix} = -9 - 10 = -19 \neq 0$$

$\Rightarrow$  system has unique solution.

$$\Delta_1 = \begin{vmatrix} -11 & 5 \\ 18 & -3 \end{vmatrix} = 33 - 90 = -57$$

$$\Delta_2 = \begin{vmatrix} 3 & -11 \\ 2 & 18 \end{vmatrix} = 54 + 22 = 76$$

$\therefore$  by Cramer's rule

$$x = \frac{\Delta_1}{\Delta} = \frac{-57}{-19} = 3, y = \frac{\Delta_2}{\Delta} = \frac{76}{-19} = -4$$

$$\therefore x = 3, y = -4$$

#### 1. (ii)

Given system of equations is

$$x - 2y = 5 \quad \dots (1)$$

$$-2x + 4y = 8 \quad \dots (2)$$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & -2 \\ -2 & 4 \end{vmatrix} = 4 - 4 = 0$$

$\therefore$  system has either no solution or infinite many solutions.

$$\Delta_1 = \begin{vmatrix} 5 & -2 \\ 8 & 4 \end{vmatrix} = 20 + 16 = 36 \neq 0$$

$$\therefore \Delta = 0 \text{ but } \Delta_1 = 36 \neq 0$$

$\Rightarrow$  system has no solution.

#### 1. (iii)

Given system of equations is

$$2x - y = 6 \quad \dots (1)$$

$$-6x + 3y = -18 \quad \dots (2)$$

$$\text{Here, } \Delta = \begin{vmatrix} 2 & -1 \\ -6 & 3 \end{vmatrix} = 6 - 6 = 0$$

$$\Delta_1 = \begin{vmatrix} 6 & -1 \\ -18 & 3 \end{vmatrix} = 18 - 18 = 0, \Delta_2 = \begin{vmatrix} 2 & 6 \\ -6 & -18 \end{vmatrix} = -36 + 36 = 0$$

$$\therefore \Delta = 0, \Delta_1 = \Delta_2 = 0$$

$\Rightarrow$  system has infinite many solutions and given by putting either x or y equal to some arbitrary constant.

Let  $y = k$ , where k is any real number.

$$\text{Equation (1)} \Rightarrow 2x = 6 + k \Rightarrow x = \frac{6+k}{2}$$

$$\therefore x = \frac{6+k}{2}, y = k, \text{ where k is any real number}$$

2,

Given system of equations is

$$x + y + 2z = 4 \quad \dots(1)$$

$$x - y + 3z = 3 \quad \dots(2)$$

$$2x + 2y + 4z = 7 \quad \dots(3)$$

$$\Delta = \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 2 & 4 \end{vmatrix}$$

Here,

Taking 2 common from  $R_3$

$$\Delta = 2 \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 1 & 2 \end{vmatrix}$$

$\left[ \because \text{if a row is multiplied with some number, the whole determinant is multiplied with that number.} \right]$

$$= 2(0) = 0$$

$[\because R_1 \text{ and } R_3 \text{ are identical}]$

$\therefore$  system has either no solution or infinite many solutions.

$$\Delta_1 = \begin{vmatrix} 4 & 1 & 2 \\ 3 & -1 & 3 \\ 7 & 2 & 4 \end{vmatrix}$$

$$\begin{aligned} \text{Expanding along } R_1 \quad \Delta_1 &= 4(-4 - 6) - 1(12 - 21) + 2(6 + 7) \\ &= -40 + 9 + 26 = -5 \neq 0 \end{aligned}$$

$\therefore$  systems have no solution.

3.

Finding  $\Delta$  :

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 2 & 9 \end{vmatrix} = -20 \neq 0$$

As  $\Delta \neq 0$ , so solution exist.

Finding  $\Delta_1, \Delta_2$ , and  $\Delta_3$  :

$$\Delta_1 = \begin{vmatrix} 6 & 2 & 3 \\ 7 & 4 & 1 \\ 14 & 2 & 9 \end{vmatrix} = -20$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 3 \\ 2 & 7 & 1 \\ 3 & 14 & 9 \end{vmatrix} = -20$$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 6 \\ 2 & 4 & 7 \\ 3 & 2 & 14 \end{vmatrix} = -20$$

$$\therefore x = \frac{\Delta_1}{\Delta} = 1, y = \frac{\Delta_2}{\Delta} = 1, z = \frac{\Delta_3}{\Delta} = 1,$$

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## 7.10 REFERENCES/ FURTHER READINGS

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1. BCS-012 : Basic Mathematics :  
[https://onlinecourses.swayam2.ac.in/nou26\\_ma07/preview](https://onlinecourses.swayam2.ac.in/nou26_ma07/preview) ( January 2026 session)
2. Mathematics Class XII Part – 1 By Ravi Prakash , GRB Books, Second Edition 2025